



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO P K AGENCIES

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

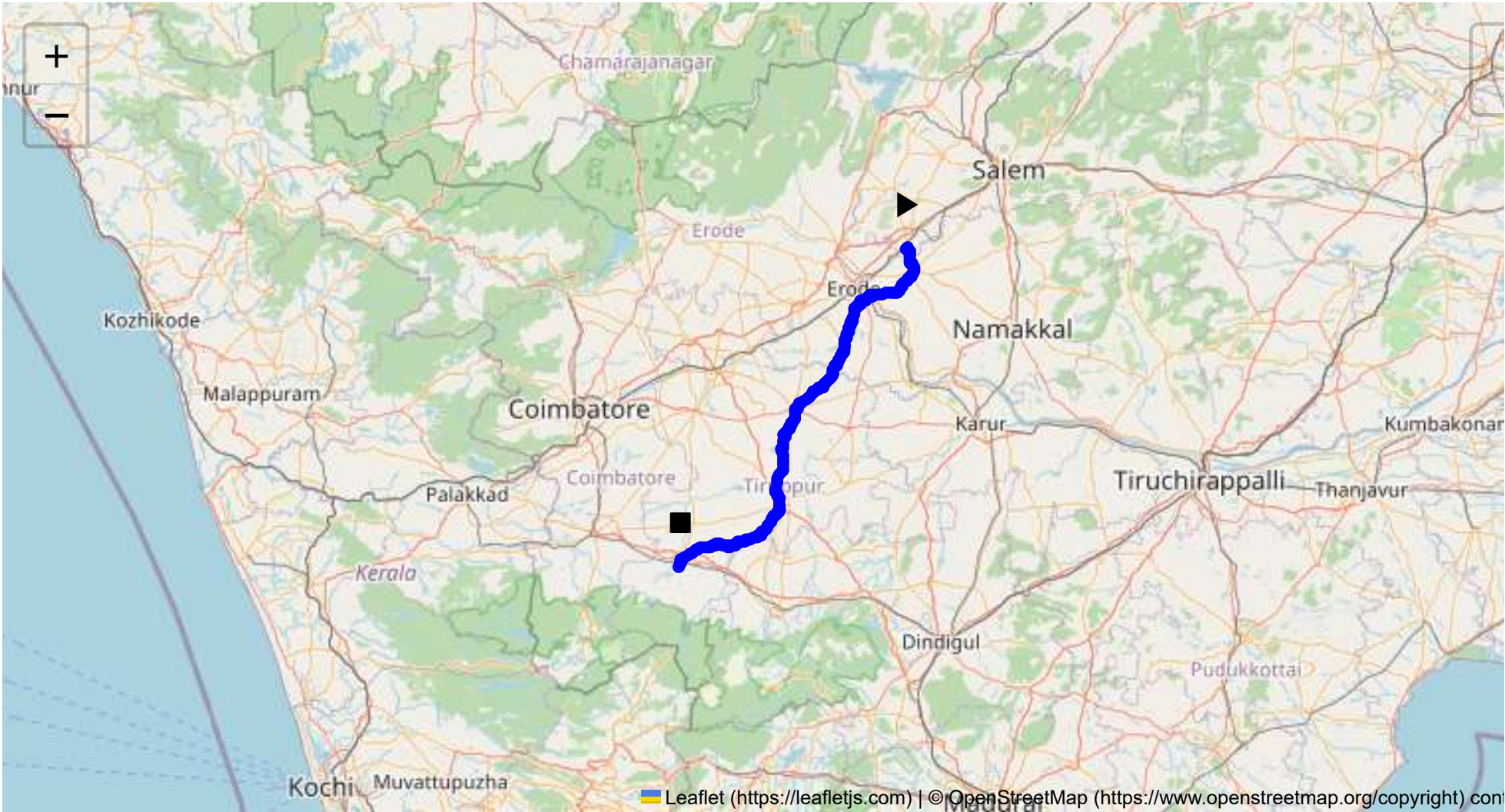
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 142.47 km**  
**Estimated Duration: 3.2 hours**  
**Adjusted Duration (Heavy Vehicle): 4.0 hours**  
**Start: (11.4381, 77.8734)**  
**End: (10.569015, 77.23829)**



**Risk Spots**

|   | RiskSpotType | Coordinates        | Speed Limit | Distance from Start |
|---|--------------|--------------------|-------------|---------------------|
| 4 | Turn         | 11.43961, 77.87341 | 15 KM/Hr    | 0.08 km             |
| 5 | Turn         | 11.44029, 77.87544 | 15 KM/Hr    | 0.28 km             |
| 6 | Turn         | 11.43111, 77.88170 | 15 KM/Hr    | 1.51 km             |



|    | RiskSpotType | Coordinates        | Speed Limit | Distance from Start |
|----|--------------|--------------------|-------------|---------------------|
| 7  | Turn         | 11.39983, 77.88377 | 15 KM/Hr    | 5.04 km             |
| 8  | Turn         | 11.39706, 77.88825 | 15 KM/Hr    | 5.62 km             |
| 0  | Roundabout   | 11.38395, 77.89480 | 15 KM/Hr    | 7.41 km             |
| 9  | Turn         | 11.37868, 77.89498 | 15 KM/Hr    | 7.78 km             |
| 10 | Turn         | 11.36602, 77.88875 | 15 KM/Hr    | 9.44 km             |
| 11 | Turn         | 11.33882, 77.86218 | 15 KM/Hr    | 13.35 km            |
| 12 | Turn         | 11.33713, 77.86222 | 15 KM/Hr    | 13.91 km            |
| 13 | Turn         | 11.31747, 77.84240 | 15 KM/Hr    | 17.10 km            |
| 14 | Turn         | 11.30951, 77.77608 | 15 KM/Hr    | 24.34 km            |
| 15 | Turn         | 11.31096, 77.77049 | 15 KM/Hr    | 25.00 km            |
| 16 | Turn         | 11.30938, 77.76674 | 15 KM/Hr    | 25.36 km            |
| 17 | Turn         | 11.29473, 77.74685 | 15 KM/Hr    | 28.33 km            |
| 18 | Turn         | 11.29133, 77.74660 | 15 KM/Hr    | 28.69 km            |
| 19 | Turn         | 11.28890, 77.74349 | 15 KM/Hr    | 29.04 km            |
| 20 | Turn         | 11.28558, 77.74271 | 15 KM/Hr    | 29.52 km            |
| 21 | Turn         | 11.27314, 77.72962 | 15 KM/Hr    | 31.54 km            |
| 22 | Blind Spot   | 11.27330, 77.72711 | 10 KM/Hr    | 31.81 km            |
| 23 | Turn         | 11.23090, 77.71808 | 15 KM/Hr    | 36.65 km            |
| 24 | Turn         | 11.15943, 77.70068 | 15 KM/Hr    | 44.89 km            |
| 25 | Turn         | 11.11943, 77.67799 | 15 KM/Hr    | 50.06 km            |
| 26 | Turn         | 11.11783, 77.67283 | 15 KM/Hr    | 50.63 km            |
| 27 | Turn         | 11.11680, 77.67167 | 15 KM/Hr    | 50.88 km            |
| 28 | Turn         | 11.10471, 77.67058 | 15 KM/Hr    | 52.22 km            |
| 29 | Turn         | 11.09508, 77.66612 | 15 KM/Hr    | 53.42 km            |
| 30 | Turn         | 11.08722, 77.66431 | 15 KM/Hr    | 54.25 km            |
| 31 | Turn         | 11.08274, 77.65832 | 15 KM/Hr    | 54.99 km            |
| 32 | Turn         | 11.06187, 77.63939 | 15 KM/Hr    | 58.33 km            |
| 33 | Turn         | 11.06173, 77.63591 | 15 KM/Hr    | 58.62 km            |
| 34 | Turn         | 11.00847, 77.56981 | 15 KM/Hr    | 67.92 km            |
| 35 | Turn         | 11.00090, 77.56888 | 15 KM/Hr    | 68.98 km            |
| 1  | Roundabout   | 11.00172, 77.56268 | 15 KM/Hr    | 69.53 km            |
| 36 | Turn         | 10.94900, 77.54681 | 15 KM/Hr    | 75.93 km            |
| 37 | Turn         | 10.94764, 77.54709 | 15 KM/Hr    | 76.19 km            |
| 38 | Turn         | 10.93853, 77.53921 | 15 KM/Hr    | 77.51 km            |

|    | RiskSpotType | Coordinates        | Speed Limit | Distance from Start |
|----|--------------|--------------------|-------------|---------------------|
| 39 | Turn         | 10.93794, 77.53764 | 15 KM/Hr    | 77.70 km            |
| 40 | Turn         | 10.93006, 77.53262 | 15 KM/Hr    | 78.73 km            |
| 41 | Turn         | 10.89465, 77.52809 | 15 KM/Hr    | 82.72 km            |
| 42 | Turn         | 10.89185, 77.52641 | 15 KM/Hr    | 83.11 km            |
| 43 | Turn         | 10.83431, 77.52934 | 15 KM/Hr    | 89.34 km            |
| 44 | Turn         | 10.77602, 77.50716 | 15 KM/Hr    | 96.66 km            |
| 45 | Turn         | 10.75794, 77.51938 | 15 KM/Hr    | 99.01 km            |
| 46 | Turn         | 10.72791, 77.52050 | 15 KM/Hr    | 102.11 km           |
| 2  | Roundabout   | 10.72668, 77.52217 | 15 KM/Hr    | 102.68 km           |
| 47 | Turn         | 10.63694, 77.40463 | 15 KM/Hr    | 119.80 km           |
| 48 | Turn         | 10.62887, 77.38647 | 15 KM/Hr    | 122.28 km           |
| 49 | Turn         | 10.62275, 77.30111 | 15 KM/Hr    | 131.92 km           |
| 3  | Roundabout   | 10.58763, 77.24722 | 15 KM/Hr    | 139.14 km           |
| 50 | Turn         | 10.58787, 77.24608 | 15 KM/Hr    | 139.27 km           |

## Emergency Services

|    | type     | name                             | coordinates            | side_of_road  |
|----|----------|----------------------------------|------------------------|---------------|
| 0  | hospital | Tiruchengode, Goverment Hospital | 11.3903328, 77.8920627 | Opposite Side |
| 1  | hospital | SPM Medical Centre,Tiruchengode  | 11.3881331, 77.8931963 | Opposite Side |
| 2  | clinic   | Kongu Nursing Home               | 11.3783065, 77.8961134 | Driver's Side |
| 3  | hospital | T.C.A Hospital Tiruchengode      | 11.3791885, 77.8965774 | Driver's Side |
| 4  | hospital | Soorya Multispecialty Hospital   | 11.3786429, 77.8931912 | Opposite Side |
| 5  | hospital | Tiruchengode Government Hospital | 11.37645, 77.89426     | Opposite Side |
| 7  | hospital | Krishna Hospital, Namakkal       | 11.3754811, 77.8931817 | Opposite Side |
| 8  | hospital | Government Hospital              | 11.2343326, 77.719226  | Driver's Side |
| 10 | police   | Kangeyam Police Station          | 11.0011411, 77.562688  | Opposite Side |
| 11 | hospital | J K Hospital                     | 11.0042298, 77.5603345 | Opposite Side |
| 15 | hospital | Sri Balaji Surya Hospital        | 10.735518, 77.5185388  | Driver's Side |
| 16 | hospital | Ramana Hospital, Dharapuram      | 10.7351092, 77.517107  | Opposite Side |
| 18 | hospital | Janani Hospital, Dharapuram      | 10.7294033, 77.5210119 | Driver's Side |
| 19 | clinic   | Punitha Nursing Home             | 10.72901, 77.520148    | Driver's Side |
| 20 | hospital | Ponnaiah Hospital, Dharapuram    | 10.72916, 77.520131    | Driver's Side |
| 21 | pharmacy | Apollo Pharmacy                  | 10.5860212, 77.2456546 | Driver's Side |

|    | type     | name                           | coordinates            | side_of_road  |
|----|----------|--------------------------------|------------------------|---------------|
| 23 | hospital | Anitha Hospital                | 10.58568, 77.248727    | Driver's Side |
| 24 | hospital | ARVI Hospital                  | 10.58528, 77.249095    | Driver's Side |
| 25 | hospital | Government Hospital, Udumalpet | 10.5836488, 77.247474  | Driver's Side |
| 26 | hospital | Dr.Sivakani Hospital Pvt Ltd   | 10.5809554, 77.2461377 | Driver's Side |

## Rest Stops

|    | type       | name                 | coordinates            | side_of_road  |
|----|------------|----------------------|------------------------|---------------|
| 6  | restaurant | Lijo                 | 11.3805115, 77.8946385 | Opposite Side |
| 9  | fuel       | Hindustan Petroleum  | 11.001711, 77.565054   | Opposite Side |
| 12 | restaurant | Hotel Surya Pure Veg | 11.0016663, 77.5638487 | Driver's Side |
| 13 | fuel       | Bharat Petroleum     | 11.0017719, 77.5629134 | Driver's Side |
| 14 | fuel       | Bharat Petroleum     | 10.9903599, 77.5638924 | Opposite Side |
| 17 | restaurant | Amaravathi Lodge     | 10.7341612, 77.5177641 | Opposite Side |
| 22 | restaurant | sri saravana         | 10.5881218, 77.2478941 | Driver's Side |

## Welcome to the Journey Risk Management Study

Certainly! Below is a detailed safety analysis of the specified route from Sangagiri to Udumalaipettai via the mentioned waypoints.

### 1. Overview of the Route Map

- The route spans approximately 142.47 kilometers, winding through urban, semi-urban, and rural areas of Tamil Nadu. The journey starts in Sangagiri and passes through Puthur, Tiruchengode, Lakkapuram, Erode, Kangeyam, Uthiyur, and finally Udumalaipettai.
- The routing involves state highways and local roads.

### 2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Climate:** Tropical, with hot summers (March to June), a monsoon season (July to September), and mild winters (November to February).
- **Hazards:** During the monsoon, heavy rains can cause slippery roads and increased chances of landslides or flooding, particularly in low-lying areas and near rivers.

### 3. Analysis of Traffic Patterns

- **Peak Hours:** Mornings (8:00-10:00 AM) and evenings (5:00-7:00 PM) see heightened traffic, especially near towns like Erode and Tiruchengode.
- **Congestion-Prone Areas:** Urban centers and school zones, particularly near Erode and Tiruchengode, often experience bottlenecks.

### 4. Assessment of Road Quality and Infrastructure

- Roads are generally well-paved along state highways but might degrade on some rural stretches. Watch for potholes, especially during post-monsoon periods.
- Road signs and lighting can be sparse in certain rural areas, increasing night-time navigation difficulty.

### 5. Suggestions for Alternative Routes for Emergencies

- **Via NH 81:** For faster travel in emergencies, consider using NH 81 connecting to nearby urban centers, though it may not pass precisely through all designated points.

### 6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permits:** Ensure that the vehicle has a permit for hazardous materials.
- **Curfews and Restrictions:** Check for any local restrictions on timing or route changes, particularly through densely populated or sensitive areas.

### 7. Overview of Historical Incidents

- Heavy vehicles carrying hazardous materials have occasionally faced mishaps, primarily due to road conditions and mechanical failures. Most incidents documented involve minor spills with no major environmental impacts.

### 8. Environmental Considerations and Sensitive Areas

- Pass near agricultural lands, requiring adherence to strict spill prevention measures to avoid contamination.
- Proximity to water bodies, particularly during monsoon, necessitates enhanced caution to prevent spillage into streams or rivers.

### 9. Analysis of Communication Coverage

- Cellular service is generally reliable but expect potential dead zones in rural areas such as Uthiyur. Equip drivers with emergency communication devices where necessary.

## 10. Estimated Emergency Response Times

- Urban areas like Erode have faster response times (20-30 minutes), while rural stretches could take significantly longer, up to 1-2 hours.

## 11. An Overall Summary of Risk Assessment

- **Low-Medium Risk:** While most of the route goes through well-trafficked and maintained areas, caution is advised during monsoon times and in lesser-developed segments. Proper pre-trip inspections and adherence to traffic norms can significantly mitigate risks.

In conclusion, the safety of this route for hazardous materials transport can be managed effectively with proper planning, adherence to regulations, and by staying informed about local weather and traffic conditions. It is imperative for drivers to be trained in emergency response and to have contingency plans and communication systems in place.

### Route Photos of Risky Spots



**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 0.28 km

**Coordinates:** 11.44029, 77.87544





**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 1.51 km**

**Coordinates: 11.43111, 77.88170**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 5.04 km**

**Coordinates: 11.39983, 77.88377**





**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 5.62 km**

**Coordinates: 11.39706, 77.88825**



**Risk Type: Roundabout**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 7.41 km**

**Coordinates: 11.38395, 77.89480**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 7.78 km**

**Coordinates: 11.37868, 77.89498**





**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 9.44 km

**Coordinates:** 11.36602, 77.88875



**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 13.35 km

**Coordinates:** 11.33882, 77.86218





**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 13.91 km**

**Coordinates: 11.33713, 77.86222**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 17.10 km**

**Coordinates: 11.31747, 77.84240**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 24.34 km**

**Coordinates: 11.30951, 77.77608**





**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 25.00 km

**Coordinates:** 11.31096, 77.77049



**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 25.36 km

**Coordinates:** 11.30938, 77.76674





**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 28.33 km**

**Coordinates: 11.29473, 77.74685**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 28.69 km**

**Coordinates: 11.29133, 77.74660**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 29.04 km**

**Coordinates: 11.28890, 77.74349**





**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 29.52 km

**Coordinates:** 11.28558, 77.74271



**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 31.54 km

**Coordinates:** 11.27314, 77.72962





**Risk Type:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Distance from Start:** 31.81 km  
**Coordinates:** 11.27330, 77.72711



**Risk Type:** Turn  
**Speed Limit:** 15 KM/Hr  
**Distance from Start:** 36.65 km  
**Coordinates:** 11.23090, 77.71808



**Risk Type:** Turn  
**Speed Limit:** 15 KM/Hr  
**Distance from Start:** 44.89 km  
**Coordinates:** 11.15943, 77.70068





**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 50.06 km

**Coordinates:** 11.11943, 77.67799



**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 50.63 km

**Coordinates:** 11.11783, 77.67283





**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 50.88 km**

**Coordinates: 11.11680, 77.67167**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 52.22 km**

**Coordinates: 11.10471, 77.67058**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 53.42 km**

**Coordinates: 11.09508, 77.66612**





**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 54.25 km

**Coordinates:** 11.08722, 77.66431



**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 54.99 km

**Coordinates:** 11.08274, 77.65832





**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 58.33 km**

**Coordinates: 11.06187, 77.63939**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 58.62 km**

**Coordinates: 11.06173, 77.63591**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 67.92 km**

**Coordinates: 11.00847, 77.56981**





**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 68.98 km

**Coordinates:** 11.00090, 77.56888



**Risk Type:** Roundabout

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 69.53 km

**Coordinates:** 11.00172, 77.56268





**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 75.93 km**

**Coordinates: 10.94900, 77.54681**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 76.19 km**

**Coordinates: 10.94764, 77.54709**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 77.51 km**

**Coordinates: 10.93853, 77.53921**





**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 77.70 km

**Coordinates:** 10.93794, 77.53764



**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 78.73 km

**Coordinates:** 10.93006, 77.53262





**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 82.72 km**

**Coordinates: 10.89465, 77.52809**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 83.11 km**

**Coordinates: 10.89185, 77.52641**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 89.34 km**

**Coordinates: 10.83431, 77.52934**





**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 96.66 km

**Coordinates:** 10.77602, 77.50716



**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 99.01 km

**Coordinates:** 10.75794, 77.51938





**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 102.11 km**

**Coordinates: 10.72791, 77.52050**



**Risk Type: Roundabout**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 102.68 km**

**Coordinates: 10.72668, 77.52217**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 119.80 km**

**Coordinates: 10.63694, 77.40463**





**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 122.28 km**

**Coordinates: 10.62887, 77.38647**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 131.92 km**

**Coordinates: 10.62275, 77.30111**





**Risk Type:** Roundabout

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 139.14 km

**Coordinates:** 10.58763, 77.24722



**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 139.27 km

**Coordinates:** 10.58787, 77.24608

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